



PORTFOLIO

NATALIE ŽIŽKOVÁ
ARCHITECTURAL DESIGN 2

ARCHIP

ARCHITECTURAL
INSTITUTE
IN PRAGUE

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01. STUDIO BRIEF

STUDIO NAME: Rethinking Logistics

PROFESORS: Šimon Prokop, Lukáš Kurilla

My studio this semester focused on rethinking logistics and exploring how warehouses could become beneficial parts of the city rather than harmful infrastructures. We worked on transforming logistics facilities into community assets through collaboration with industry partners such as Ramboll, Henning Larsen, and Panattoni. Throughout the semester, I engaged with evidence-based design using real performance data and applied parametric evaluation methods to link design decisions with measurable outcomes such as energy use, biodiversity, stormwater management, carbon footprint, and cost. I also explored the balance between regulatory constraints and regenerative design ambitions while working in a collaborative, interdisciplinary environment with engineers, consultants, and developers.

The studio emphasized real-world impact, as our proposals were developed for actual industry conditions rather than fictional scenarios. My work contributed to discussions presented to professionals and stakeholders, with the potential to be exhibited, published, and included in a broader catalogue of strategies aimed at shaping future logistics development across Europe. The main task was to develop and defend a specific design strategy—ranging from peripheral high-performance facilities to integrated urban hybrids—supported by data-driven arguments rather than purely visual representations. Overall, the studio provided practical experience in professional workflows, teamwork, and communication, while addressing one of the key contemporary challenges in sustainable urban development.



02. LOCATION

CITY LOCATION: Lakeview village, Mississauga, Ontario, Canada

LOGICTIS CENTER LOCATION: 1260 Lakeshore Rd E, Mississauga, ON L5E 1E9, Kanada

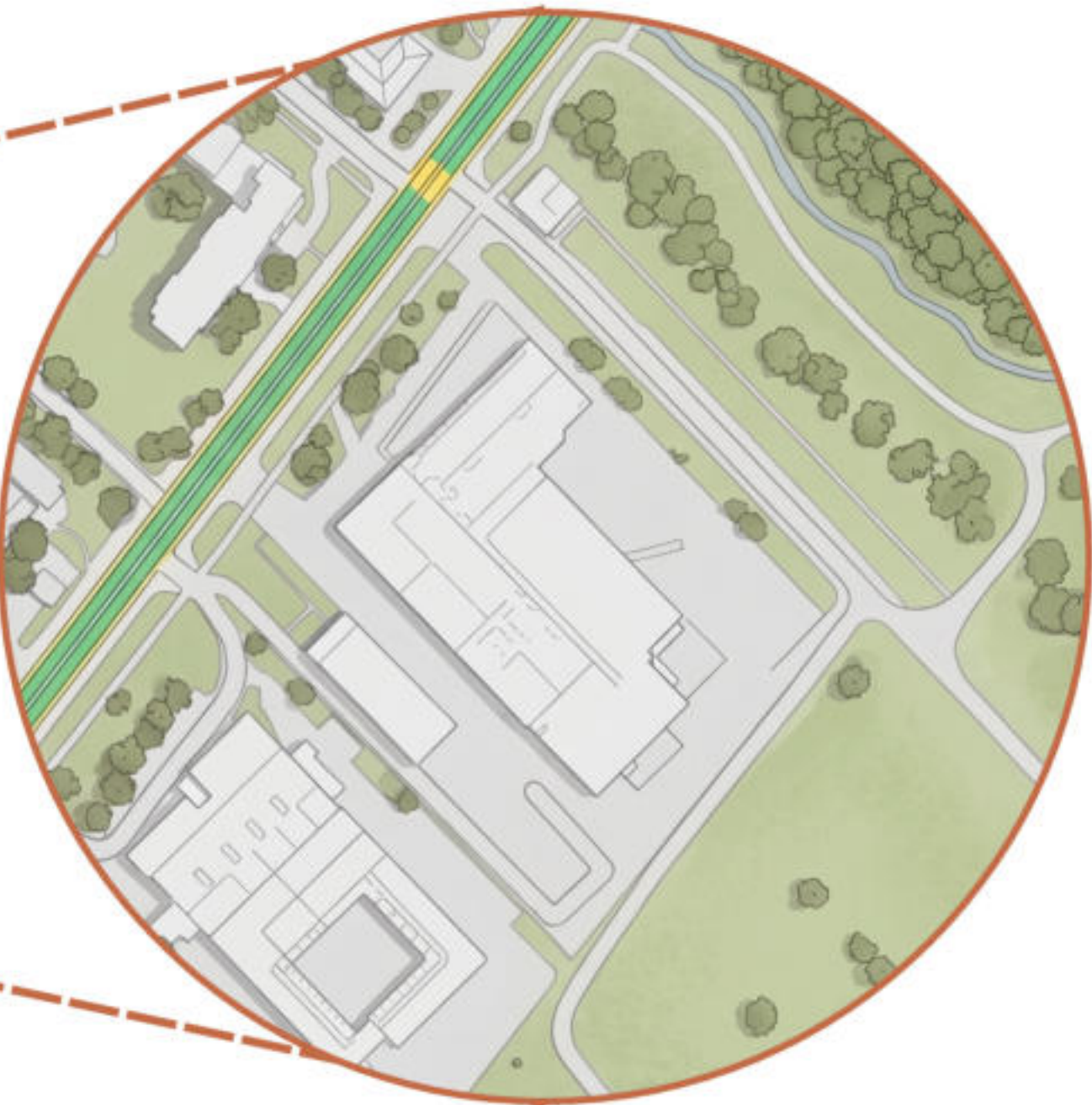
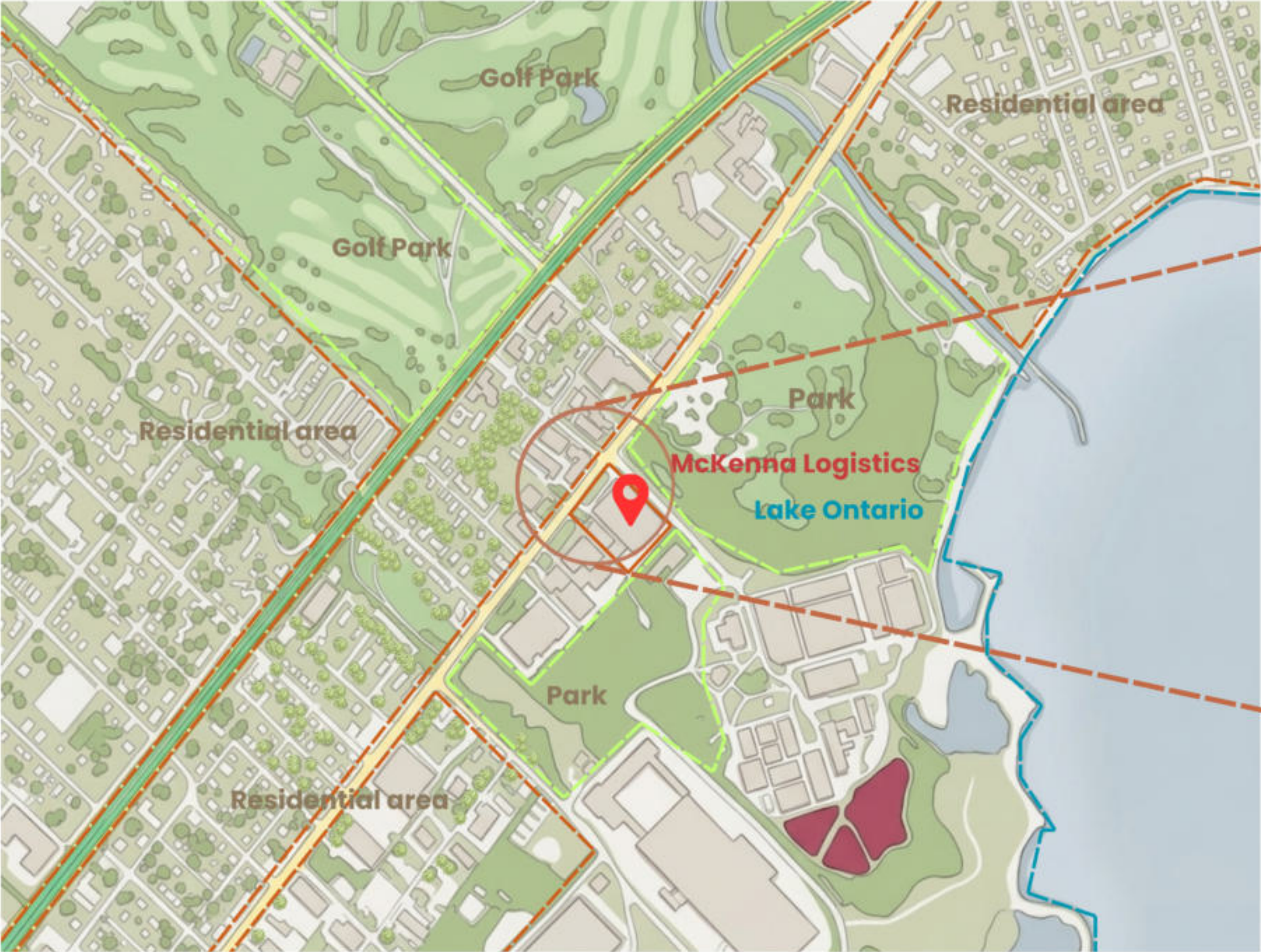
The Story of Lakeview Village

My project is located in Lakeview Village in Mississauga, Ontario, just a few miles away from the busy city of Toronto. Situated right on the beautiful shores of Lake Ontario, this area is currently going through one of the biggest urban transformations in the region. What is now becoming a green, luxury neighborhood has a surprisingly heavy industrial past.

For decades, this exact site was defined by a massive coal-fired power station. The most famous part of this plant were its four huge chimneys, which the locals naturally called the "Four Sisters." These towers completely dominated the skyline and served as a symbol of the area's industrial era. However, as the city shifted its focus toward a greener and cleaner future, the power plant was shut down, and the iconic Four Sisters were eventually demolished to make space for something new. Today, the former industrial zone is being completely rebuilt from the ground up. The empty fields and old factory grounds are turning into a high-end residential district filled with modern apartments, houses, and expansive public parks. The goal is to create a peaceful, nature-focused community right by the water. However, right in the middle of this beautiful new development sits an active logistics center that still holds onto that cold, industrial feeling. It disrupts the new residential vibe and breaks the connection between the parks—which is exactly the challenge my project sets out to solve.



SITE ANALYSIS



McKenna Logistics
1260 Lakeshore Rd E, Mississauga

Lakeview village, Mississauga, Ontario, Canada

FLOW ANALYSIS



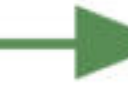
Cars



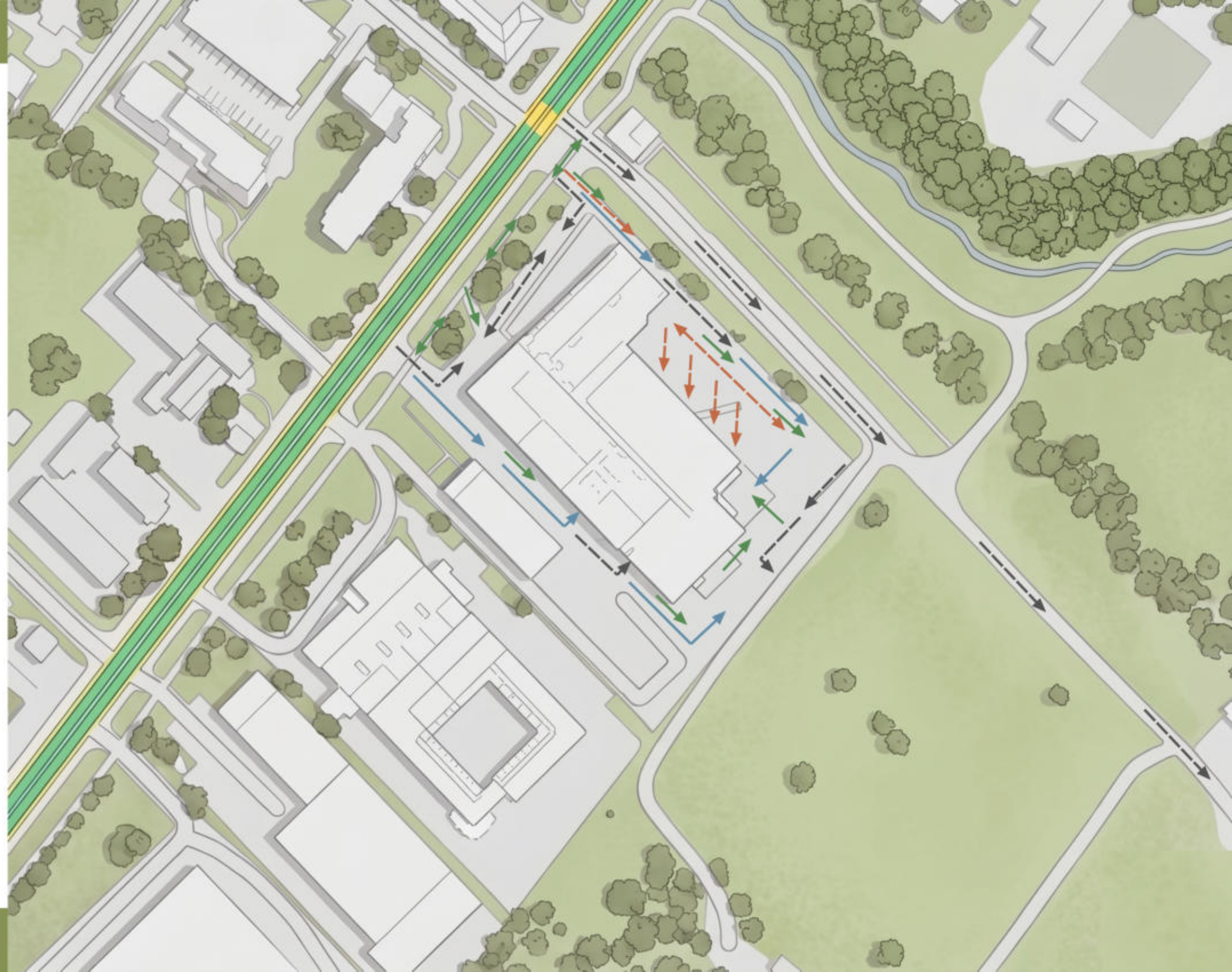
Trucks



Staff-McKenna



Public



PROBLEMS

1. Unsafe Public Access

Visitors walking to the physiotherapy clinic must cross the heavy truck route. This is very dangerous and creates a high risk of collisions.

2. Dangerous Parking Route

Staff arriving by car have to drive through the main road used by heavy trucks. This makes reaching the staff parking difficult and unsafe.

3. Risky Walk to the Office

After parking, employees still have to cross the busy paths of both cars and trucks just to enter their office building safely.

4. Disconnected Parks

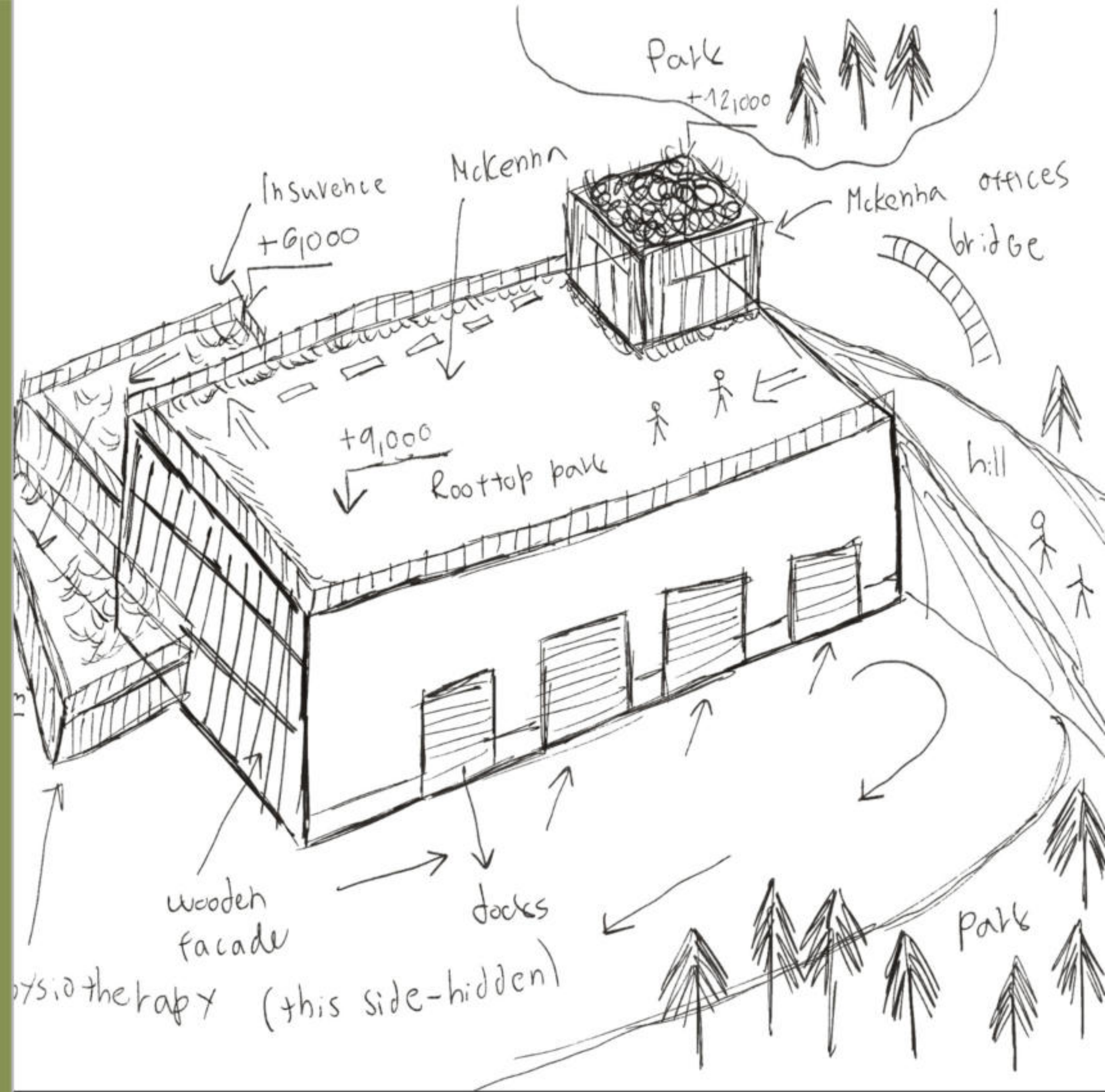
Two public parks are separated by a busy road leading to the coast. This physical barrier completely breaks the natural connection between the green areas.



03. CONCEPT Invisible Logistics

The Concept: Invisible Logistics

The fundamental concept driving my design is the idea of "Invisible Logistics." Instead of merely trying to soften the harsh appearance of an industrial building, my ultimate goal was to make it completely dissolve into the surrounding urban fabric. While traditional logistics centers create physical and visual barriers, my approach relies on architectural camouflage to heal the broken structure of the city. I aimed to entirely suppress the monumental scale of the facility, shifting the focus away from the industrial machinery and back to the human experience. The primary objective is for the building to silently operate as a crucial logistical engine on the inside, while its true function remains entirely concealed from the public eye. Ultimately, my vision was to design a space that no longer reads as a logistics hub at all, but simply exists as a natural, inviting extension of the community's parks and residential life.



04. STRATEGY CCC Method

The CCC Method Strategy

When I started working on a solution for this site, I realized I needed a clear system to tackle all its complex problems. That is why I created a strategy that I named the 3C Method. I named it this way because my entire approach stands on three main actions: Conceal, Connect, and Coexist.

- **CONCEAL:** First, I needed to hide the massive industrial building. By covering it with a green hill and wrapping it in a warm wooden facade, I managed to conceal the logistics center so it no longer looks like a harsh factory to the public.
- **CONNECT:** Second, the original site acted as a physical barrier. My goal was to connect the two previously separated parks using a new pedestrian tunnel, a bridge, and a walkable roof, turning the area back into a continuous green space.
- **COEXIST:** Finally, because it is still an active logistics hub, safety was key. By dividing the area into strict, separate zones for heavy trucks, staff, and visitors, I made sure that the heavy industry and everyday community life can safely coexist in the exact same location.

1. CONCEAL

1. SKETCH: NATURE+LOGISTICS



By blending the surrounding nature with the harsh industrial volume, I transformed the rigid box into an organic extension of the landscape. The green roof and the artificial hill completely hide the true scale of the facility.

2. SKETCH: BEFORE



In its original state, the stark, grey industrial facade aggressively disrupted the warm residential streetscape. It stood out as an unwelcome concrete barrier that simply did not belong in the neighborhood.

3. SKETCH: AFTER

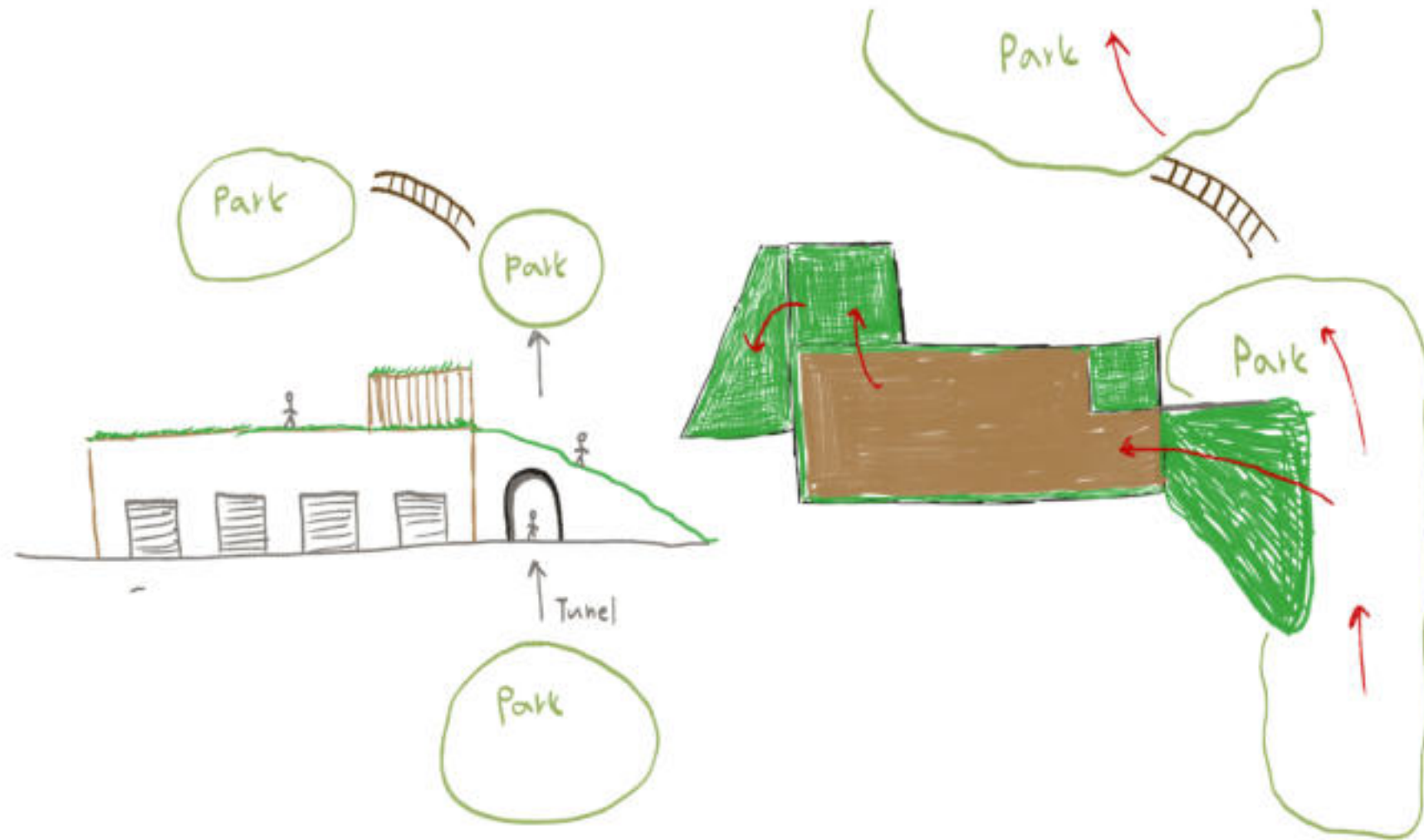


By wrapping the center in a warm timber facade, the building now perfectly matches the local residential aesthetic. This architectural camouflage allows the logistics hub to blend in so naturally that it becomes almost invisible.

2.CONNECT

1.GREEN ROOFS+PEDESTRIAN TUNEL

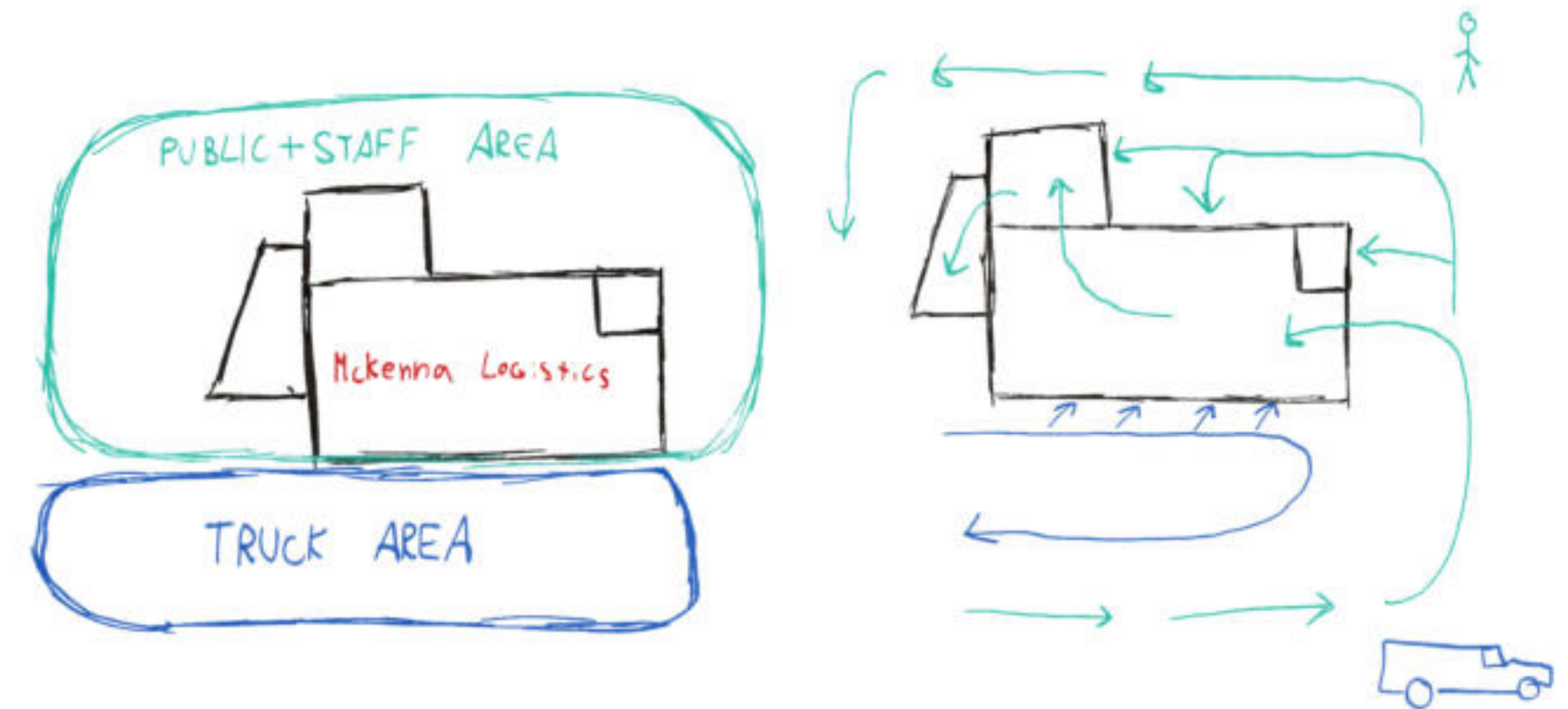
The artificial hill and green roof create a continuous, walkable pathway that naturally links the surrounding green spaces."A safe pedestrian tunnel runs directly underneath the landscaped hill to connect the divided park zones.



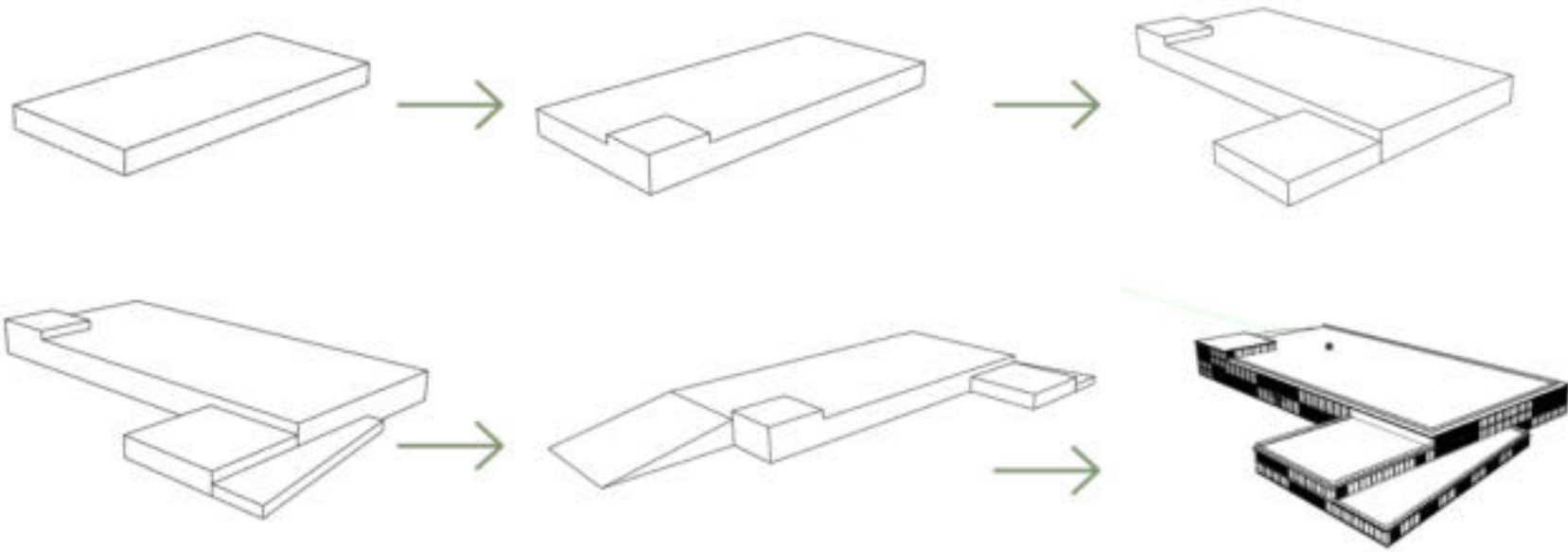
3.COEXIST

ZONING AND TRAFFIC FLOWS

The site is strictly divided into two distinct zones to keep heavy trucks completely separated from the public and staff. Pedestrian and vehicle routes never cross, ensuring that heavy logistics and everyday community life can safely coexist.

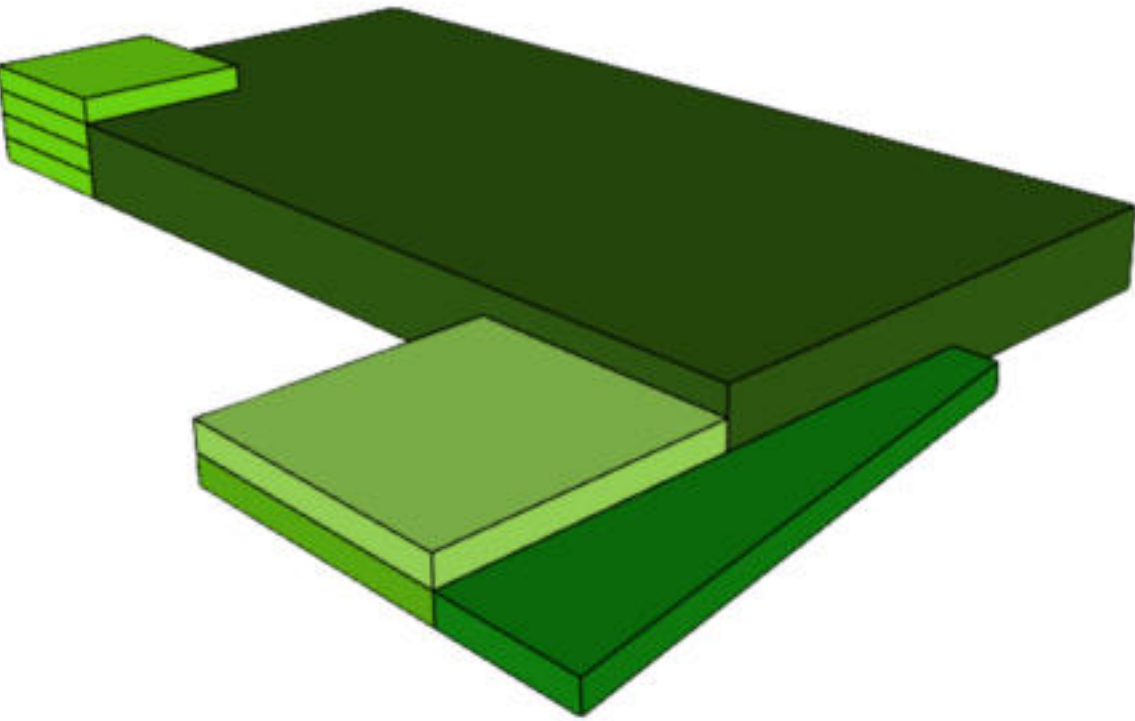


FORM DIAGRAM



PROGRAM DIAGRAM

- Insurence
- McKenna offices
- Physiotherapy
- Offices
- Logistic centre

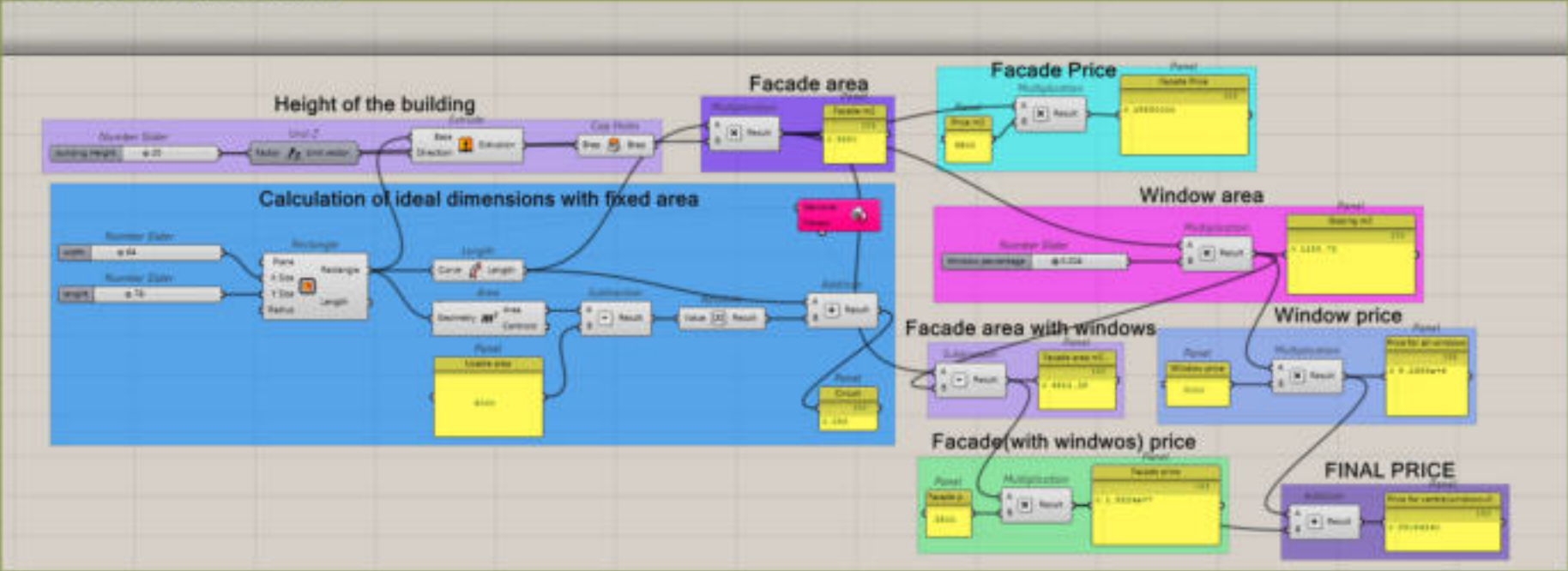


05. GEOMETRY OPTIMIZER

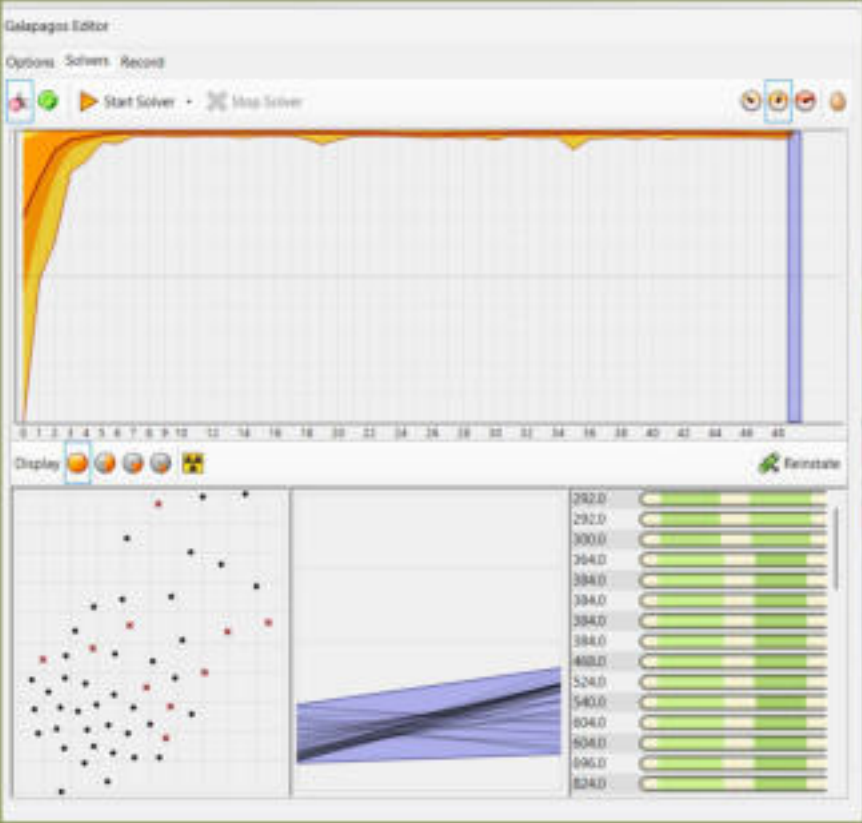
Geometry Optimization

To find the most efficient dimensions for the logistics center, I developed a geometry optimizer using Grasshopper. The primary goal of the script is to generate building shapes that strictly maintain a required usable area of 5,000 m² while minimizing the total perimeter. This optimization is crucial, as it significantly reduces facade construction costs and minimizes operational energy loss through the building's envelope. The evolutionary solver operates on a specific fitness function: minimizing the total length of the walls while applying a mathematical penalty for any deviation from the 5,000 m² target area. Beyond spatial optimization, the script also functions as a real-time financial tool. As the geometry changes, it automatically calculates essential data, including the total facade area, the required window area, and their respective material costs. Ultimately, it instantly generates the final estimated price for the entire building envelope, allowing me to make design decisions that are both architecturally and economically sustainable.

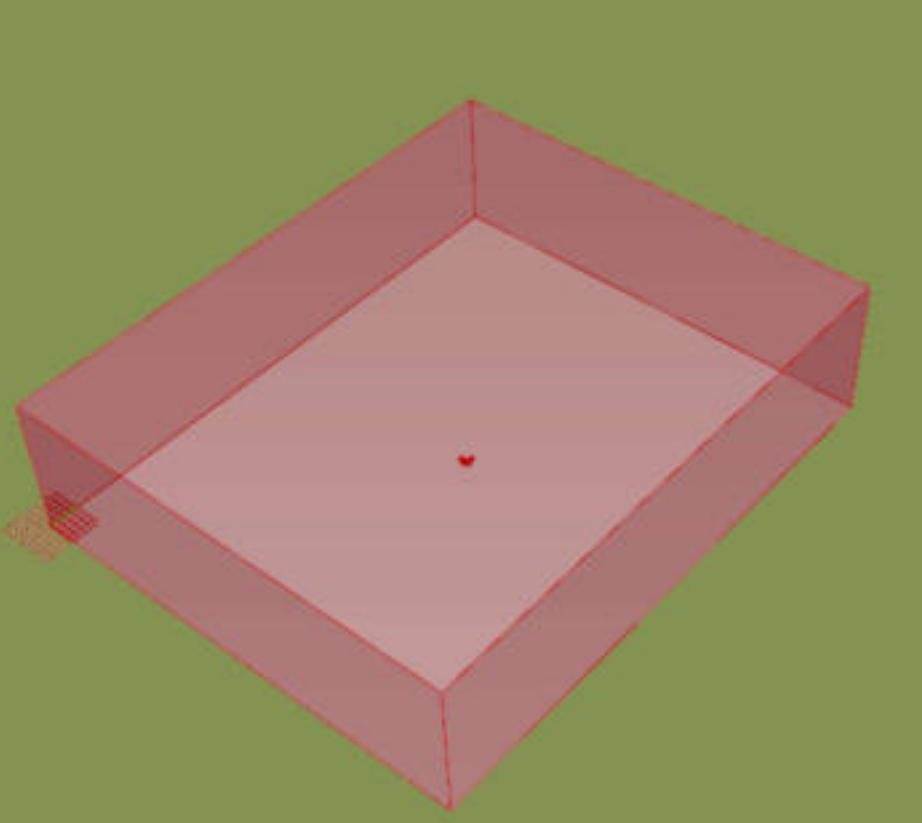
GRASSHOPPER-SCRIPT



GRAF-CALCULATING



3D MODEL SHOWING FINAL RESULT



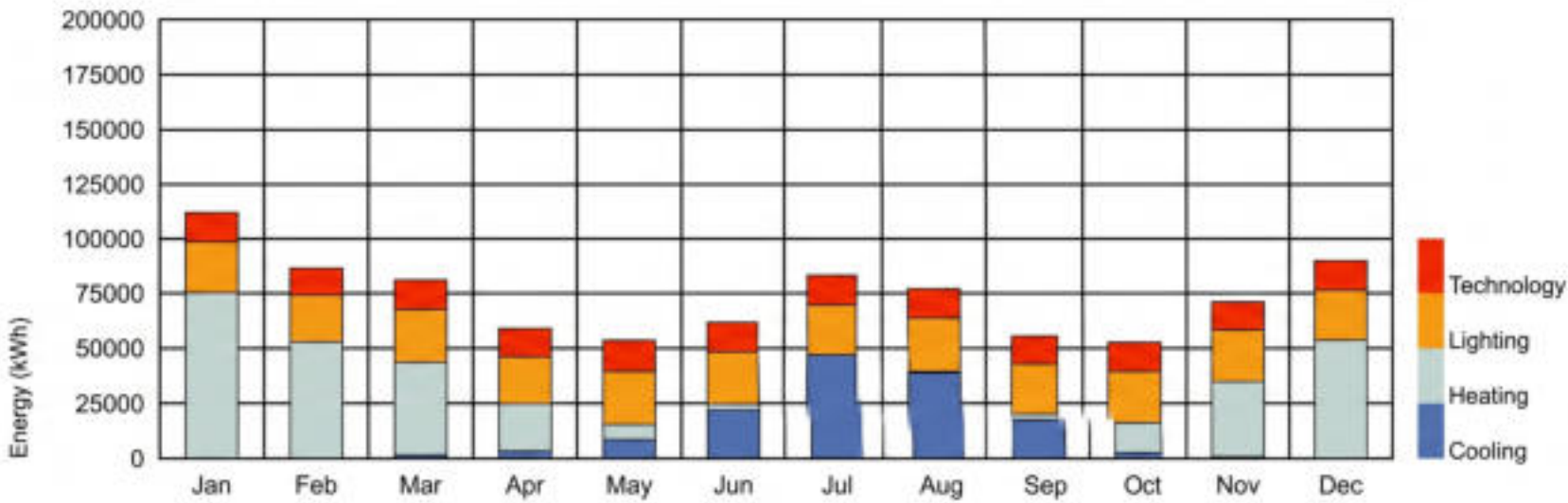
06.

ENERGY PERFORMANCE

ENERGY PERFORMANCE SIMULATION

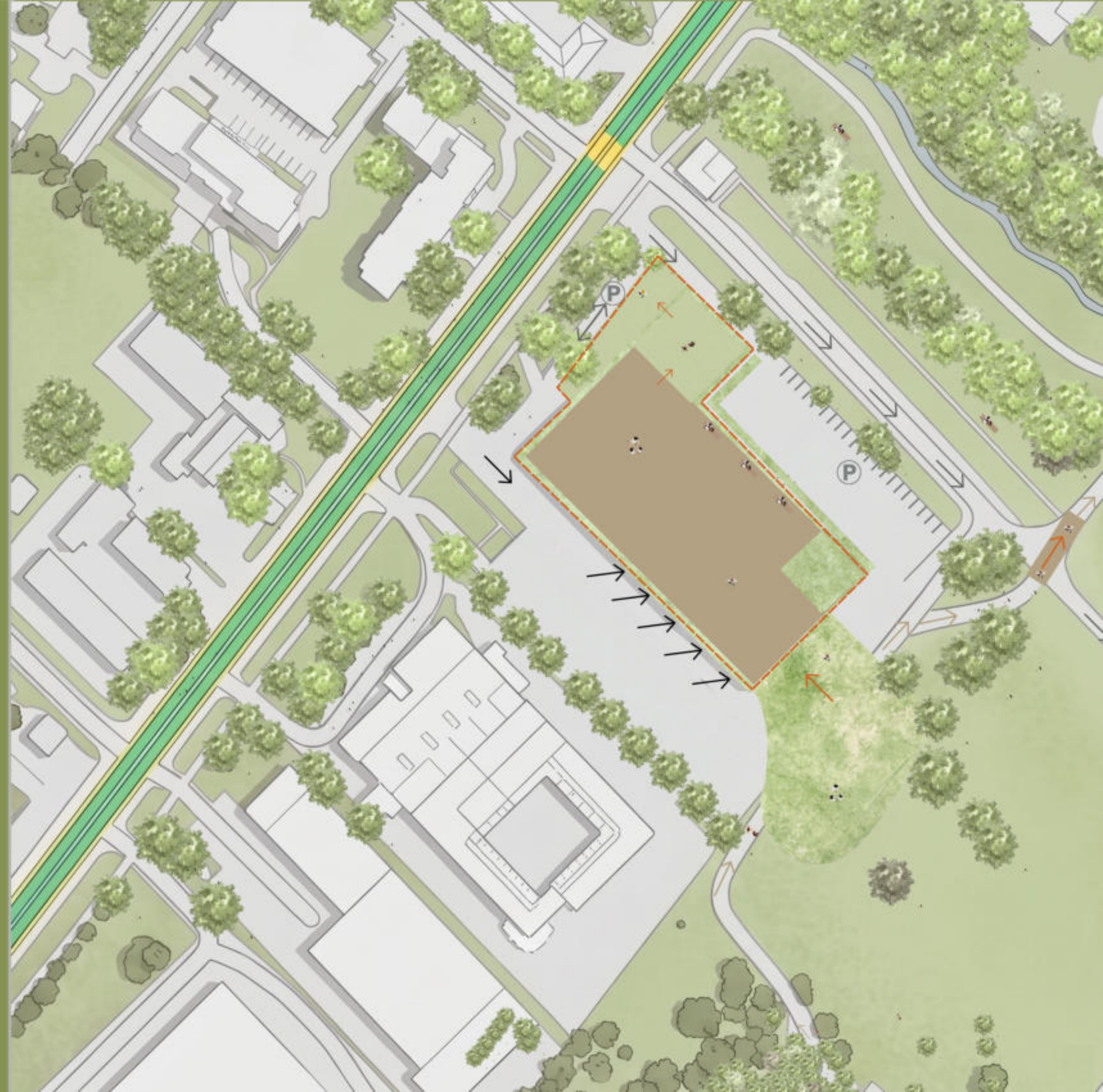
To evaluate the environmental footprint and overall efficiency of the logistics center, I used a Grasshopper script to run a comprehensive energy simulation. The resulting graph illustrates the building's estimated monthly energy consumption in kilowatt-hours (kWh) throughout an entire year. It clearly breaks down the total energy demand into four key operational categories: Heating, Cooling, Lighting, and Technology.

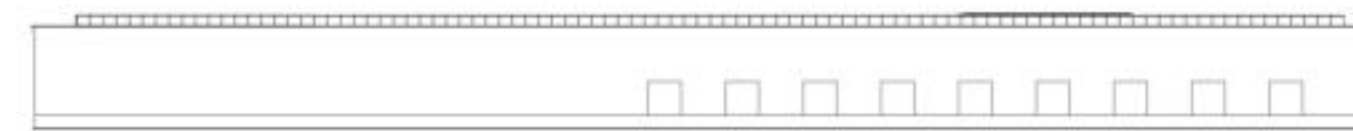
As the data shows, the building's energy profile responds naturally to the local Canadian climate. Heating demands predictably peak during the cold winter months (January, February, December), while cooling becomes necessary during the summer peak. Meanwhile, the energy required for lighting and technology remains relatively stable year-round. Conducting this parametric analysis allowed me to understand the true operational cost of the building and verify that the proposed design is not only spatially optimized but also highly energy-efficient.



07. DESIGN

My design focuses on practical architectural solutions to fix the site's main problems. First, to stop the dangerous mix of heavy trucks and pedestrians, I divided the area into strictly separated zones. Second, to soften the harsh industrial look, the whole building is now wrapped in a warm timber facade, making it feel much more welcoming. The biggest intervention is the artificial green hill that covers the facility and leads visitors up to the roof level. However, the roof itself is designed as a cascading landscape with different levels. The highest point of the building is not walkable; instead, it is completely covered by an intensive green roof to maximize nature. From the highest walkable level, which is surrounded by greenery, visitors can step down to two lower, accessible roof terraces, each dropping by three meters and featuring fully green, usable spaces. Underneath this landscape, I designed a pedestrian tunnel that leads to a new bridge, safely connecting the two separated parks. Together, these architectural elements transform the building from a concrete block into an active, multi-level part of the landscape.



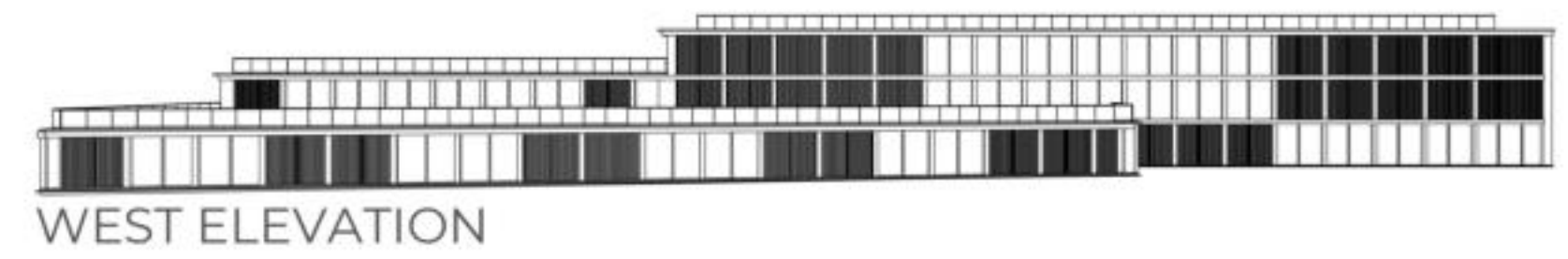


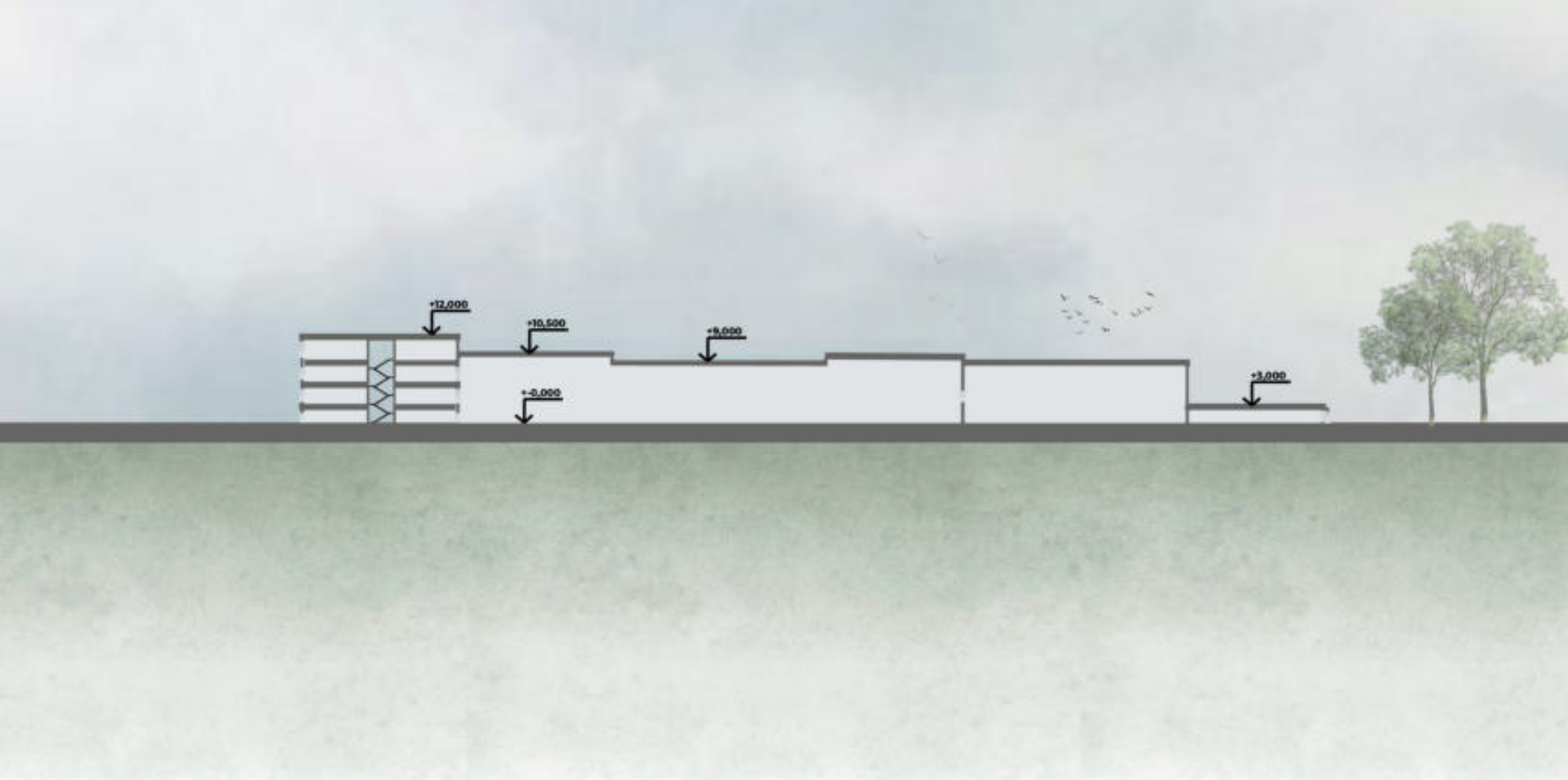
SOUTH ELEVATION



NORTH ELEVATION

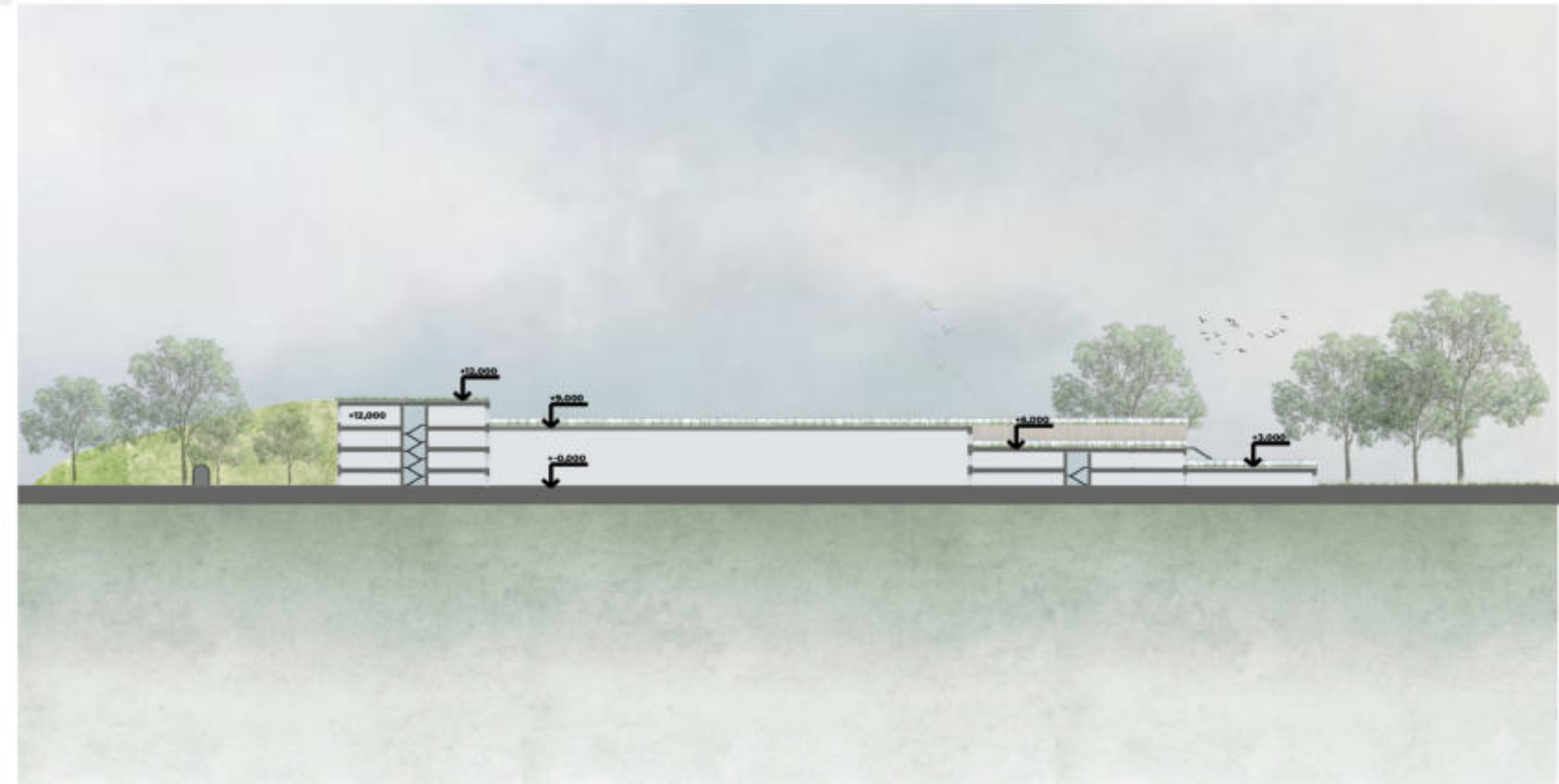






SECTION A-A BEFORE

SECTION A-A AFTER



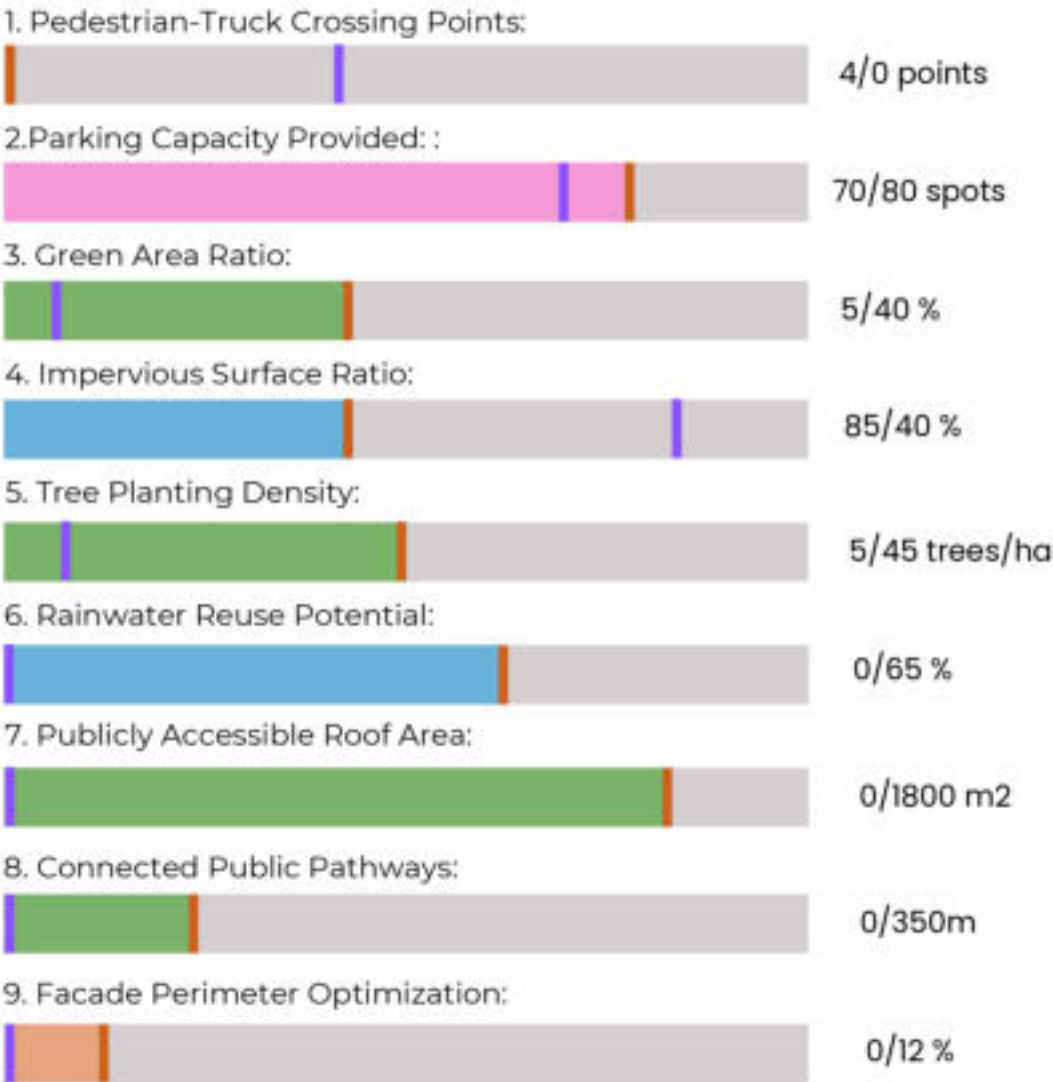






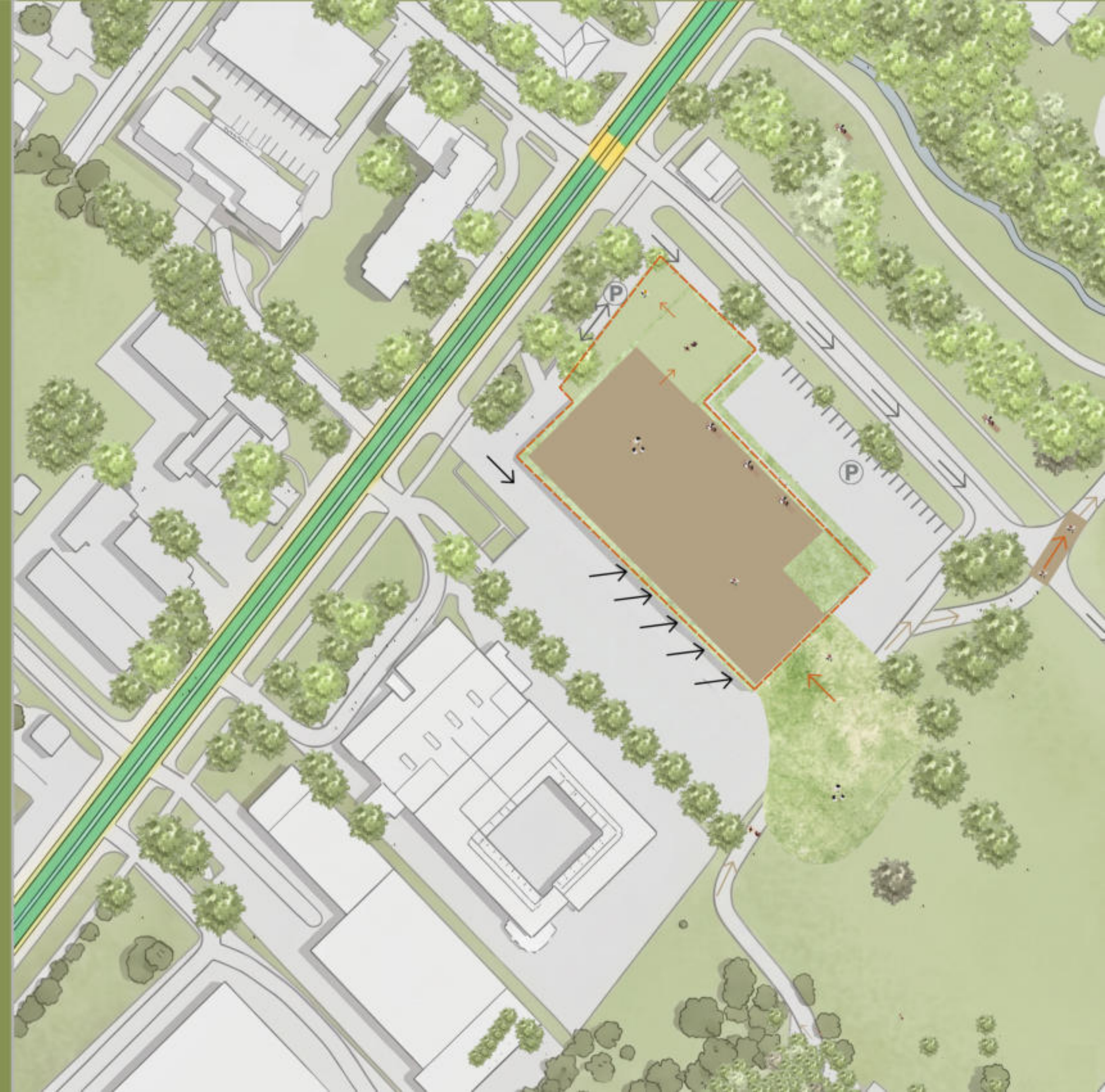
08. KEY PERFORMANCE INDICATORS

To ensure the proposed design truly solves the site's challenges, I used a set of Key Performance Indicators (KPIs) to compare the original state with the new architectural intervention. The resulting data clearly demonstrates the success of my 3C Method. Most importantly, public safety is now guaranteed by reducing the dangerous pedestrian-truck crossing points to zero, without slowing down the daily logistics. Environmentally, the massive increase in the Green Area Ratio and the new tree planting density prove that the industrial footprint has been successfully returned to nature. Ultimately, these metrics prove that the concept of "Invisible Logistics" is not just a nice aesthetic vision, but a highly functional, safe, and sustainable reality.



09. CONCLUSION

By redesigning this problematic logistics hub, the project unlocks the site's true potential and gives something valuable back to the community. What used to be an ugly barrier is now an interesting piece of architecture that fits nicely into both the residential area and the nature around it. Because of the public green spaces, the building actually improves the neighborhood instead of just taking up space. At first glance, you wouldn't even realize it's a logistics center because it looks like a natural part of the landscape. In the end, it proves that with good design, industry and everyday life can exist together perfectly.



10. APPENDIX

